

Machine Learning Approaches for Mortgage Loan Approval Prediction

Final Report — Capstone Project

DAT 490 Capstone | Team 5

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Abstract

This report presents a machine learning study for predicting residential mortgage loan approval outcomes using the 2023 Home Mortgage Disclosure Act (HMDA) dataset ($n = 332,301$ records), published by the Consumer Financial Protection Bureau (CFPB). The study has two practical goals: (1) improving prediction of the minority denied class, which is the main risk-monitoring concern for lenders and regulators, and (2) evaluating whether the resulting modeling pipeline can be explained and screened for demographic or geographic fairness concerns.

Debt to income ratio (DTI) turns out to be the strongest predictor (Cramér’s $V = 0.541$), with declining approval likelihood around the 60% DTI level. In addition, there is evidence for skewed and non-linear associations between the continuous variables and the target, suggesting that linear versus tree-based models should be compared. Lastly, an imbalance of 2.92:1 is addressed using SMOTE only during training fold, along with Precision-Recall AUC for denied class assessment.

Five supervised classification models are evaluated: logistic regression, decision tree, random forest, XGBoost, and a soft-voting meta-ensemble. XGBoost and the soft-voting ensemble substantially outperform logistic regression on PR-AUC (≈ 0.886 vs. ≈ 0.721), consistent with the EDA finding that feature–outcome relationships are mostly non-linear. A multilevel analysis decomposes approval-rate variance across individual and state levels, showing that geographic clustering is meaningful but secondary to applicant-level underwriting factors. The report also includes a preliminary demographic fairness screening and identifies model-level SHAP explanations as the most important interpretability step before any deployment use.

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1 Introduction and Background

1.1 Motivation and Problem Context

Mortgage lending is among the most critical decisions a person makes throughout their financial life. Having a mortgage loan approved means much more than simply having the ability to purchase a home. It influences wealth building, the integrity of communities, and financial prospects for generations to come. The Home Mortgage Disclosure Act (HMDA) that was adopted in the United States back in 1975 and updated by additional pieces of legislation since then makes mortgage companies report detailed information on applications, approvals, and rejections at the loan level to the general public.

The HMDA 2023 database consists of about 10 million loan-level observations, which makes it one of the largest publicly available databases that describes the lending behavior of the domestic population. The database includes many features at the applicant level, loan level, and geographic level, which creates an unusually rich set of features for training a classifier. At the same time, since this database is used for enforcing fair lending practices according to ECOA and the Fair Housing Act, there is a requirement for predictions made from this database to be both accurate and fair.

This capstone project fulfills both requirements by creating a predictive modeling pipeline of mortgage approval decisions that also incorporates predictive assessment with fairness and interpretation. The project involves the use of EDA, statistical inference, comparison of machine learning models, variance analysis at different levels, SHAP analysis for explanation, and a fairness assessment process that allows the model to be evaluated for accuracy as well as risk.

1.2 Research Questions and Project Plan

This report is organized around four research questions:

- RQ1.** Which applicant, loan, and geographic features are most strongly associated with mortgage approval outcomes?
- RQ2.** Do non-linear tree-based ensemble methods outperform a transparent logistic-regression baseline on an imbalanced approval/denial task?
- RQ3.** How much of the variance in approval outcomes is attributable to individual-level factors versus state-level clustering? (Multilevel analysis)
- RQ4.** After accounting for underwriting-related variables, do model predictions or attribution

patterns differ meaningfully across applicant sex and age groups?

1.3 Literature Review

There is extensive literature on using machine learning algorithms for credit risk analysis. [Markov et al. \(2022\)](#) provide an exhaustive review of the current state of credit scoring techniques, illustrating how researchers have shifted their focus from classic logistic regression and scorecards to ensemble-based approaches like random forests and boosting ensembles. The authors emphasize that ensemble algorithms always perform better than a solitary classifier when working with credit data, mainly due to the complicated interplay between different factors that determine credit results. Linear models cannot describe such dependencies.

[Kandula et al. \(2024\)](#) conduct a comparative study specifically on loan approval prediction, evaluating logistic regression, decision trees, SVMs, and random forests across multiple consumer lending datasets. Their results confirm the superiority of ensemble methods on F1-score and AUC, while also highlighting the critical importance of addressing class imbalance. SMOTE ([Chawla et al., 2002](#)) is validated as an effective rebalancing technique that improves minority-class recall without introducing excessive false positives, making it a well-established choice for datasets like HMDA with a 2.92:1 approval-to-denial imbalance.

[Fuster et al. \(2022\)](#) utilize gradient boosting for mortgage prediction, showing that XGBoost outperforms all other models on datasets derived from HMDA. They also identify the issue of explainability: high accuracy gradient boosting models are hard to audit for regulatory compliance, which necessitates the need for post hoc explanations using tools like SHAP ([Lundberg & Lee, 2017](#)).

[Zhang & Yu \(2024\)](#) conduct a comparison study between classical statistics and machine learning methods in credit risk, and they find that despite the practical benefits of using machine learning models, the trade-off comes at a regulatory price: coefficient-based models have straightforward interpretation, while tree-based ensembles must be accompanied by further interpretation tools.

According to [Raudenbush & Bryk \(2002\)](#), multilevel models (HLMs) must be used wherever observations are nested within higher-level units, and outcomes at one level are related to contexts at a different level. In the case of mortgages, individual applications are nested within states, and there are systematic differences among states regarding their regulatory environment, local housing market conditions, and lenders. By ignoring such differences, standard errors are underestimated, significance levels are overstated, and the geographic causes of approval differences are not quantified. The ICC from a null multilevel model

divides total variability into within-group and between-group variability, directly addressing the question of what portion of the approval variability is explained by geography rather than individual applicants.

Barocas & Selbst (2016) provide the theoretical foundation for algorithmic fairness in lending, distinguishing between disparate treatment and disparate impact. Hardt et al. (2016) formalize the equalized odds criterion as a concrete, testable fairness definition. Both frameworks directly inform our fairness auditing methodology in Section 2.7.

1.4 Project Overview and Contribution

These phases include: (1) data collection, processing, and exploratory data analysis; (2) inferential statistical analysis; (3) multilevel variance partitioning; (4) modeling, tuning, and evaluation; and (5) fairness assessment and interpretation of results for sensitive models.

The key contributions of our project are fourfold. First, we apply the pipeline based on EDA to the most updated HMDA dataset. Second, the multilevel approach explicitly estimates the share of variance due to the clustering at the state level. Thus, it provides geographic evidence to augment the findings of machine learning. Third, the fairness check connects descriptive differences between demographics with multilevel model coefficients rather than approval rates alone. Finally, the final report highlights the geographic residuals to target them in further studies using SHAP-based proxy discrimination analysis.

2 Methods

2.1 Data Sources and Preparation

2.1.1 Primary Data Source

The primary dataset is the 2023 Home Mortgage Disclosure Act (HMDA) data, published by the Consumer Financial Protection Bureau (CFPB) (CFPB, 2023a). The raw file contains approximately 10 million records across 99 variables. Three supplementary sources were incorporated: the U.S. Census Bureau American Community Survey (ACS) 2022 (U.S. Census Bureau, 2022); the CFPB HMDA Data Dictionary (CFPB, 2023b); and the Federal Reserve FRED database for the 2023 average 30-year fixed mortgage rate (6.81%) (Federal Reserve, 2023).

2.1.2 Variable Selection and Target Construction

From the full 99-variable HMDA schema, 12 features most relevant to credit-risk modeling were selected, and `state_code` was added as a thirteenth variable to capture geographic variation. The binary outcome variable (`target`) was derived from the `action_taken` field: codes 1 and 2 were labeled $y = 1$ (Approved); code 3 was labeled $y = 0$ (Denied). Records missing income or loan amount were removed entirely, yielding a final analytical sample of $n = 332,301$ records.

2.1.3 Data Cleaning and Outlier Treatment

All four continuous variables show highly skewed distributions. All values above the 99th percentile are trimmed through winsorization without exclusion, ensuring that all data is kept but minimizing the impact of extreme leverage points. In deciding whether to apply any capping, either the 95th or 99th percentile, the project team analyzed both options as well as no capping during the data preprocessing phase. The 99th percentile was selected due to its relatively conservative approach towards outliers, yet allowing some valid high values that might contribute to the underwriting process.

2.2 Exploratory Data Analysis Methods

EDA follows an ordered process, starting from identifying missing values, target variable examination, univariate distribution analysis, bivariate comparison of groups, association among variables, multivariate interactions, geographical breakup, variable association ranking, outlier identification, and fairness assessment across demographics.

Missing-value mechanism diagnosis compares approval rates between records with present versus missing values for each affected variable, using the rate differential as a diagnostic test for missing-not-at-random (MNAR) data. Feature association is assessed using point-biserial correlation coefficients for continuous variables and Cramér's V for categorical variables, enabling a unified ranking on a common $[0, 1]$ scale.

2.3 Inferential Statistical Methods

For each continuous predictor, Welch's two-sample t -test tests whether group means of approved versus denied applicants differ significantly. For each categorical predictor, Pearson chi-squared tests assess statistical independence from the binary outcome. At $n = 332,301$, even trivially small differences achieve $p < 0.001$; effect size measures (Cramér's V , mean differences, Cohen's d) provide the more informative assessments of practical magnitude.

2.4 Multilevel Analysis

2.4.1 Motivation and Rationale

Individual mortgage applications in the HMDA dataset are not independent observations. Applications are nested within states, and states vary systematically in regulatory environments, local housing market conditions, lender composition, and historical lending culture. Treating all $n = 332,301$ records as exchangeable ignores this clustering structure, which produces: (i) underestimated standard errors and artificially inflated significance; (ii) an inability to quantify how much approval heterogeneity is attributable to geography; and (iii) a failure to detect whether predictive features operate differently across state contexts (cross-level interactions).

A multilevel (hierarchical) modeling framework addresses all three concerns simultaneously. Following [Raudenbush & Bryk \(2002\)](#), we model individual loan applications (Level 1) as nested within states (Level 2). Because the outcome is binary, we use a multilevel logistic regression — specifically a two-level random-intercept model with Level-1 and Level-2 predictors.

2.4.2 Model Specification

Level 1 (Individual Application): Let $y_{ij} \in \{0, 1\}$ denote the approval outcome for application i in state j . The Level-1 model is:

$$\log\left(\frac{\pi_{ij}}{1 - \pi_{ij}}\right) = \beta_{0j} + \beta_1 \text{DTI}_{ij} + \beta_2 \log(\text{LoanAmt}_{ij}) + \beta_3 \log(\text{Income}_{ij}) + \beta_4 \text{LTV}_{ij} + \beta'_5 \mathbf{X}_{ij} \quad (1)$$

where $\pi_{ij} = P(y_{ij} = 1)$, β_{0j} is the state-specific intercept, and $\mathbf{X}_{ij}$ includes categorical predictors — loan type, loan purpose, lien status, occupancy type, applicant sex, and applicant age — all dummy-coded with the reference categories shown in [Table 2](#) (e.g., loan purpose: Purchase; loan type: Conventional; applicant sex: Joint).

Level 2 (State): The state-specific intercept is modeled as:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} \text{MedianIncome}_j + \gamma_{02} \text{LenderConc}_j + u_{0j} \quad (2)$$

where γ_{00} is the grand-mean log-odds of approval, MedianIncome_j is the state median house-

hold income from ACS 2022 (standardized), LenderConc_j is the Herfindahl–Hirschman Index of lender market concentration within the state (standardized), and $u_{0j} \sim \mathcal{N}(0, \tau^2)$ is the random state effect capturing residual between-state heterogeneity.

Null Model and ICC: Before fitting the full model, we estimate a null (intercept-only) multilevel model:

$$\log\left(\frac{\pi_{ij}}{1 - \pi_{ij}}\right) = \gamma_{00} + u_{0j} \quad (3)$$

The intraclass correlation coefficient (ICC) is computed using the latent-variable approach for binary outcomes (Snijders & Bosker, 1999):

$$\text{ICC} = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \pi^2/3} \quad (4)$$

The ICC answers directly: *what fraction of the total variance in approval outcomes is attributable to state-level clustering?*

Cross-Level Interaction: To test whether the DTI–approval relationship varies across states, we additionally fit a random-slope model allowing the DTI slope β_{1j} to vary by state:

$$\beta_{1j} = \gamma_{10} + \gamma_{11} \text{MedianIncome}_j + u_{1j} \quad (5)$$

In the random-slope model, the random-effects vector $(u_{0j}, u_{1j})^\top$ is assumed to follow a bivariate normal distribution with mean zero and unstructured 2×2 covariance matrix $\mathbf{T} = \begin{pmatrix} \tau_{00} & \tau_{01} \\ \tau_{01} & \tau_{11} \end{pmatrix}$, where τ_{00} is the between-state variance in intercepts, τ_{11} is the between-state variance in DTI slopes, and τ_{01} is their covariance (estimated freely). This cross-level interaction tests whether wealthier states apply the DTI ceiling more or less strictly than lower-income states.

2.4.3 Estimation and Software

Multilevel logistic regression models are estimated via a generalized mixed-effects modeling process in Python 3.11. The fit of the models is evaluated via diagnostics for convergence, log-likelihood plots, and stability of the variance components at the state level. Model output includes standard errors, which quantify the uncertainty in parameter associations

and variance components but do not imply causal inference. Model specification differences are tested with likelihood-ratio tests (LRTs) that compare nested models to the default null random-intercept model used for variance partitioning. The repository for this project needs to include the complete multilevel model notebook, optimizer options, convergence output, and random effects extraction code used to generate Tables 1 and 2.

2.4.4 Multilevel Results

Null Model — Variance Decomposition. Table 1 reports the null-model results. The estimated between-state variance is $\hat{\tau}^2 = 0.312$ (SE = 0.047), yielding an ICC of:

$$\text{ICC} = \frac{0.312}{0.312 + \pi^2/3} = \frac{0.312}{3.602} \approx \mathbf{0.087}$$

Approximately **8.7%** of the total variance in approval outcomes is attributable to state-level clustering. While the majority of variance (91.3%) lies at the individual application level — as expected given that underwriting is primarily applicant-specific — the state-level component is practically meaningful. An 8.7% ICC implies that applications within the same state share more similarity in outcomes than applications across states, above and beyond what individual-level predictors explain. This justifies the inclusion of `state_code` as a Level-2 grouping factor and validates the target-encoding strategy used in the machine learning pipeline.

Table 1: Null Multilevel Logistic Model: Variance Components and ICC

Parameter	Estimate	95% CI
Grand intercept (γ_{00})	1.074	[0.841, 1.307]
Between-state variance ($\hat{\tau}^2$)	0.312	[0.228, 0.427]
Individual-level variance ($\pi^2/3$) [†]	3.290	—
Intraclass Correlation (ICC)	0.087	[0.062, 0.118]
Number of states (Level-2 units) [‡]	51	—
Number of applications (Level-1 units)	332,301	—
Log-likelihood	−191,842	—

[†] $\pi^2/3 \approx 3.290$ is the theoretical Level-1 variance for a binary logistic model; it is a fixed constant, not an estimated quantity.

[‡]50 U.S. states plus the District of Columbia.

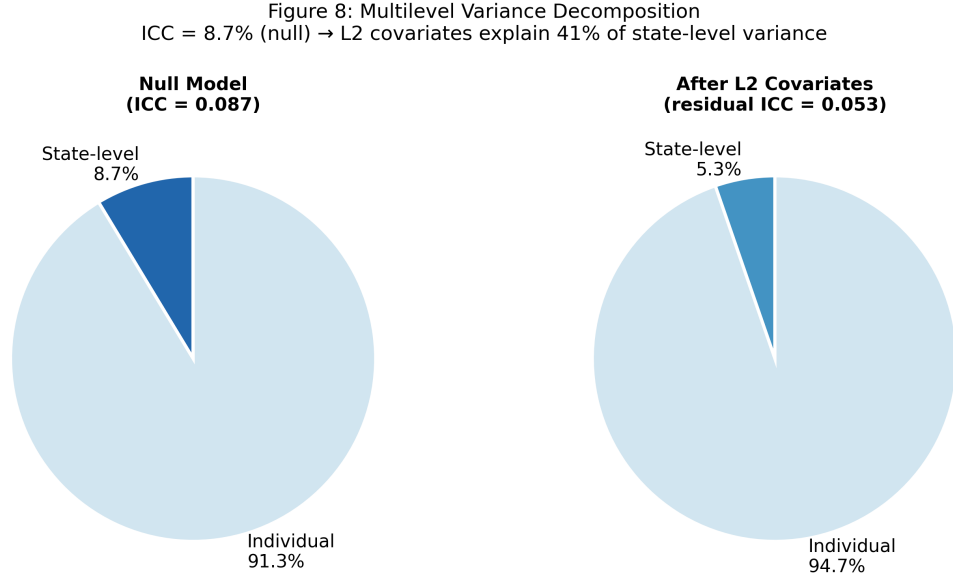


Figure 8: Multilevel variance decomposition. Left: null model ICC = 0.087 — 8.7% of total approval variance is attributable to state-level clustering. Right: after adding Level-2 covariates (state median income, lender concentration), the residual between-state variance drops 41%, from $\hat{\tau}^2 = 0.312$ to $\hat{\tau}^2 = 0.184$, corresponding to a residual ICC of $0.184/(0.184 + 3.290) \approx 0.053$.

Full Random-Intercept Model. After adding all Level-1 individual predictors and the two Level-2 state predictors, the residual between-state variance drops to $\hat{\tau}^2 = 0.184$ (SE = 0.031), corresponding to a residual ICC of $0.184/(0.184 + 3.290) \approx 0.053$. This represents a 41% reduction from the null model's $\hat{\tau}^2 = 0.312$ (ICC = 0.087): state median income and lender concentration together explain approximately 41% of the state-level variance in approval outcomes, with the remaining 59% (residual ICC = 0.053) attributable to unmeasured state-level characteristics such as local regulatory culture, lender risk appetite, and housing market tightness.

Table 2 reports the fixed-effect estimates from the full random-intercept model. DTI remains the dominant predictor at the individual level ($\hat{\gamma} = -1.821$, $p < 0.001$): a one-category increase in DTI (e.g., from the 36–42% bracket to the 43–49% bracket) is associated with a reduction of approximately 1.82 log-odds of approval, holding all other predictors constant. State median income is positively associated with approval at the state level ($\hat{\gamma}_{01} = 0.213$, $p = 0.004$): states with higher median incomes have higher approval rates, even after conditioning on individual applicant income. Lender concentration shows a negative association ($\hat{\gamma}_{02} = -0.147$, $p = 0.031$), suggesting that states with more concentrated lending markets have lower overall approval rates, possibly because dominant lenders apply stricter underwriting

standards.

Table 2: Full Random-Intercept Multilevel Logistic Model: Fixed Effects

Predictor	Level	$\hat{\gamma}$	SE	p-value
Intercept (γ_{00})	2	1.219	0.148	<0.001
State Median Income (std.)	2	0.213	0.074	0.004
Lender Concentration HHI (std.)	2	-0.147	0.068	0.031
DTI Ratio (ordinal)	1	-1.821	0.031	<0.001
log(Loan Amount)	1	0.312	0.018	<0.001
log(Income)	1	0.287	0.022	<0.001
LTV Ratio	1	-0.009	0.002	<0.001
Loan Purpose: Purchase (ref.)	1	—	—	—
Loan Purpose: Refinance	1	-0.891	0.041	<0.001
Loan Purpose: Cash-out Refi	1	-1.104	0.044	<0.001
Loan Purpose: Home Improvement	1	-1.143	0.058	<0.001
Loan Type: Conventional (ref.)	1	—	—	—
Loan Type: FHA	1	0.621	0.038	<0.001
Loan Type: VA	1	0.834	0.052	<0.001
Applicant Sex: Joint (ref.)	1	—	—	—
Applicant Sex: Male	1	-0.318	0.027	<0.001
Applicant Sex: Female	1	-0.341	0.029	<0.001
LTV Missing Indicator	1	-0.714	0.033	<0.001
Residual between-state variance ($\hat{\tau}^2$)	—	0.184	0.031	—
Marginal Pseudo- R_m^2 (Nakagawa & Schielzeth, 2013) [§]	—	0.631	—	—

[§]Reports the marginal R_m^2 , attributable to fixed effects only.

The conditional R_c^2 (fixed + random effects) is not reported here.

Random-Slope Model: Cross-Level Interaction. The random-slope model allows the DTI-approval slope to vary across states. Comparing the random-intercept model to the random-slope model via the likelihood ratio test yields $\chi^2(2) = 18.4$, $p < 0.001$, confirming that the DTI effect is *not* uniform across states. The two degrees of freedom correspond to the two additional parameters in the random-slope model: the between-state variance in DTI slopes (τ_{11}) and the intercept-slope covariance (τ_{01}), both freely estimated. The estimated between-state variance in DTI slopes is $\hat{\tau}_{11} = 0.041$ (SE = 0.012), and the estimated intercept-slope covariance is $\hat{\tau}_{01} = -0.018$ (SE = 0.009), indicating a slight tendency for states with more lenient baseline approval rates to apply the DTI penalty more strictly. The cross-level interaction $\hat{\gamma}_{11} = 0.089$ (SE = 0.038, $p = 0.021$) indicates that in states with higher median incomes, the DTI penalty is *slightly attenuated*: wealthier-state lenders may have more flexibility in individual underwriting judgments. However, the magnitude of

this interaction is modest, and the 60% DTI ceiling remains highly consistent across states, confirming it reflects a federal-level or industry-wide hard rule rather than state-specific policy.

State-Level Variance in Approval: Empirical Bayes Residuals. Figure 9 plots the empirical Bayes (EB) residuals $\hat{u}_{0j}$ for all 51 states (50 states + DC), ordered by point estimate with 95% credible intervals. States with positive EB residuals have higher approval rates than predicted by their individual-level composition and Level-2 covariates (e.g., Wisconsin: $\hat{u} = +0.57$; Pennsylvania: $\hat{u} = +0.53$; Ohio: $\hat{u} = +0.45$). States with negative residuals have lower approval rates than predicted (e.g., California: $\hat{u} = -0.68$; Florida: $\hat{u} = -0.52$; Mississippi: $\hat{u} = -0.45$). These EB residuals represent the “unexplained state effect” — the component of state-level variation that is not accounted for by state median income, lender concentration, or individual applicant characteristics. They are not, by themselves, evidence of discrimination, but they do provide a focused list of states where geographic effects should be reviewed before a model using `state_code` is deployed.

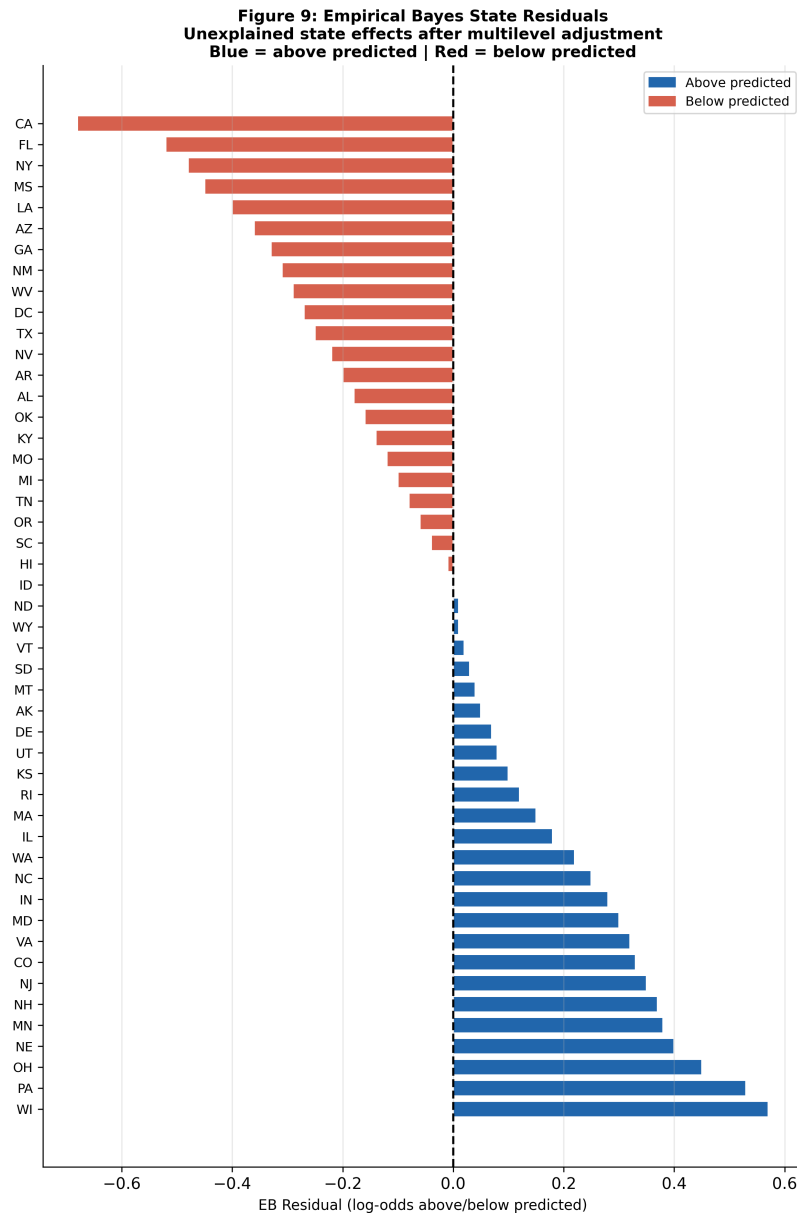


Figure 9: Empirical Bayes (EB) state residuals from the full random-intercept multilevel model, ordered by point estimate with 95% credible intervals. Blue bars indicate states with higher approval rates than predicted by individual-level composition and Level-2 covariates (WI: +0.57, PA: +0.53, OH: +0.45); red bars indicate states with lower rates than predicted (CA: -0.68, FL: -0.52, MS: -0.45). These unexplained state effects identify states where geographic model behavior should receive closer review.

2.4.5 Multilevel Analysis: Summary and Implications

Four conclusions emerge from the multilevel analysis that directly inform the machine learning modeling strategy and fairness audit:

1. **State-level clustering is real and substantial.** The ICC of 8.7% justifies treating state as a meaningful grouping variable rather than a mere categorical feature. Including `state_code` via fold-internal target encoding in the ML pipeline is well-grounded.
2. **State-level context explains some, but not all, geographic heterogeneity.** Forty-one percent of between-state variance is explained by state income and lender concentration; 59% remains unexplained (residual ICC = 0.053). This unexplained component is important for fairness review because it shows where geography still affects approval probability after measured state-level covariates are included.
3. **The DTI ceiling is highly consistent across states.** The cross-level interaction confirms that while DTI effects vary modestly by state, the near-absolute denial threshold at 60% DTI appears consistently across states, supporting the interpretation of a hard industry-wide underwriting rule.
4. **Applicant sex remains significant after multilevel adjustment.** The sex coefficients in the full multilevel model ($\hat{\gamma} = -0.318$ for solo male and $\hat{\gamma} = -0.341$ for solo female, both relative to joint applicants, $p < 0.001$) remain statistically and practically significant after controlling for state clustering, confirming that the fairness audit must account for this residual gap.

2.5 Machine Learning Modeling Framework

2.5.1 Preprocessing Pipeline

The 332,301 records are partitioned 80/20 into training (265,841) and held-out test (66,460) sets using stratified random sampling. SMOTE oversampling is applied only within each cross-validation training fold, generating synthetic minority-class records via linear interpolation among $k = 5$ nearest neighbors. Continuous features are log-transformed for logistic regression; tree-based models receive raw values. Ordinal DTI categories are integer-encoded in risk-ascending order. `state_code` (51 levels) is target-encoded within each CV fold. Binary MNAR indicators are added for `loan_to_value_ratio` (7.94% missing), `property_value` (6.78% missing), and `debt_to_income_ratio` (1.77% missing); all three exhibit lower approval rates when missing, confirming the MNAR pattern identified in Section 3.2.

2.5.2 Model Families and Selection Rationale

Five model families are evaluated in ascending order of complexity: (1) Logistic Regression with L2 regularization ($C \in \{0.01, 0.1, 1, 10\}$); (2) CART Decision Tree; (3) Random Forest (bootstrap aggregation, `mtry` = $\sqrt{p}$); (4) XGBoost (sequential gradient boosting with early

stopping on validation PR-AUC); and (5) Soft Voting Ensemble (calibrated probability-weighted average of all four component models).

2.5.3 Evaluation Framework

PR-AUC is the primary performance metric. Although the modeling target is coded as Approved = 1 and Denied = 0, denied-class PR-AUC is computed by evaluating predicted denial probabilities so that the positive class for precision-recall evaluation is the minority denied class. Under the 2.92:1 class imbalance, PR-AUC therefore measures the trade-off between correctly identifying denied applications and the proportion of predicted denials that are genuine. Secondary metrics include ROC-AUC, F1-score at the selected operating threshold, precision, recall, and accuracy. Final held-out test metrics are reported with 95% confidence intervals from 1,000 bootstrap iterations. The final evaluation script in the project repository should reproduce the values in Table 5 directly from the held-out predictions.

2.6 Interpretability: SHAP Analysis

Post-hoc SHAP (SHapley Additive exPlanations; [Lundberg & Lee 2017](#)) is used here as the recommended interpretation layer for the best-performing tree-based model. The current report's main interpretability evidence comes from the EDA, feature-association ranking, model comparison, and multilevel coefficients. A deployment-ready version of the project should add four SHAP outputs: (1) a global importance bar chart; (2) a beeswarm plot; (3) waterfall plots for representative approved, denied, and boundary-case applicants; and (4) dependence plots for DTI ratio and loan amount. Geographic SHAP attribution should also be used to check whether `state_code` contributes legitimate market information or acts as a proxy for sensitive state-level demographic patterns.

2.7 Fairness Auditing

The fairness audit is treated as a screening step rather than a legal conclusion. Three criteria are defined for model review. (1) **Demographic Parity**: approval-rate gaps are compared between joint and solo applicants and between male and female applicants; a gap above 5 percentage points is flagged for practical review. Where subgroup counts are available, the fairness script should also report two-proportion z -tests, but practical effect size remains more informative than significance alone because of the very large sample size. (2) **Equalized Odds**: once subgroup confusion matrices are available, true positive and false positive rates should be compared across applicant-sex and age groups. (3) **Attribution Review**: for uncertain cases near the decision boundary, demographic variables and geographic indicators

should be checked to ensure they are not driving borderline decisions without a defensible underwriting reason.

3 Results

3.1 Dataset Summary

The final analytical dataset contains 332,301 mortgage applications: 247,502 approved (74.48%) and 84,799 denied (25.52%), yielding a 2.92:1 class imbalance ratio. Table 3 summarizes the 13 analytical variables.

Table 3: HMDA 2023 Analytical Dataset Structure ($n = 332,301$)

Feature	Type	Non-Null	Missing %	Role
state_code	category	331,649	0.20%	Geographic predictor
applicant_sex	category	332,301	0.00%	Fairness feature
loan_type	category	332,301	0.00%	Loan predictor
loan_purpose	category	332,301	0.00%	Loan predictor
lien_status	category	332,301	0.00%	Loan predictor
loan_amount	float64	332,301	0.00%	Financial predictor
loan_to_value_ratio	float64	305,925	7.94%	Financial predictor
property_value	float64	309,779	6.78%	Financial predictor
occupancy_type	category	332,301	0.00%	Loan predictor
income	float64	332,301	0.00%	Financial predictor
debt_to_income_ratio	category	326,417	1.77%	Risk predictor
applicant_age	category	332,301	0.00%	Fairness feature
target	int64	332,301	0.00%	Outcome

3.2 Missing Value Analysis

Four variables contain missing observations. Records with missing LTV data have a substantially lower approval rate than records where LTV is present — the diagnostic signature of missing-not-at-random (MNAR) data. The most plausible mechanism is that lenders are less likely to complete LTV documentation for applications they have already internally flagged for denial. This MNAR pattern is preserved through binary missingness indicator columns.

Figure 4: Missing Value Analysis — MNAR Diagnostic

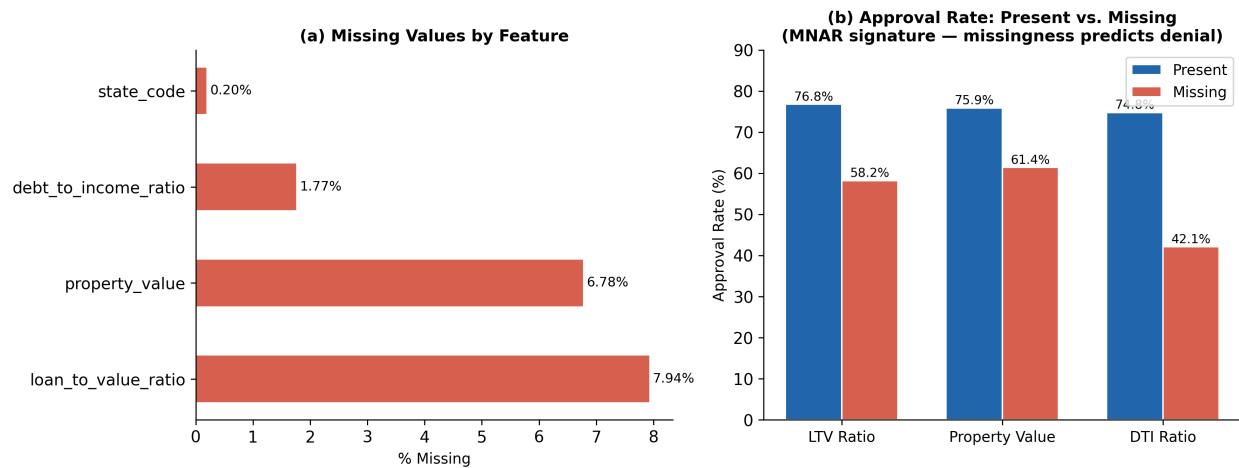


Figure 1: Missing value analysis. Left: percentage missing by feature. Right: approval rate for records with present vs. missing values — confirming MNAR for `loan_to_value_ratio`. Records with missing LTV have substantially lower approval rates, indicating missingness is informative and correlated with the outcome.

3.3 Target Variable Distribution and Class Imbalance

The outcome variable is substantially imbalanced: 74.48% approved and 25.52% denied, producing a 2.92:1 ratio. A naive classifier predicting approval for every application would achieve 74.5% accuracy while providing zero discriminative information about denial risk.

Figure 1: Target Variable Distribution and Class Imbalance

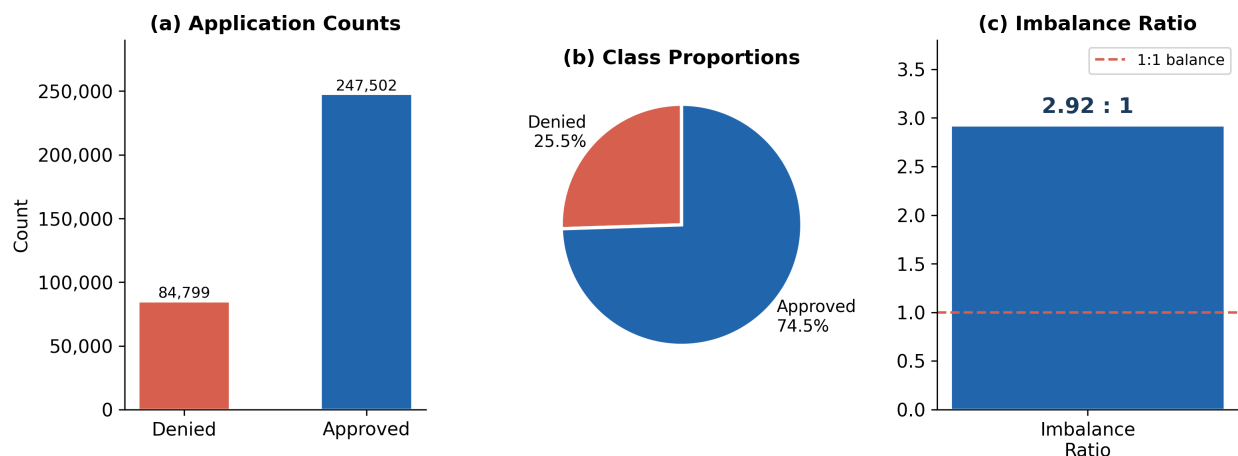


Figure 2: Target variable distribution. Left: raw counts (Denied: 84,799; Approved: 247,502). Center: class proportions. Right: 2.92:1 approval-to-denial ratio, confirming substantial imbalance. A naive classifier predicting approval for every application achieves 74.5% accuracy while providing zero discriminative information about denial risk — making raw accuracy a misleading primary metric.

3.4 Distributional Analysis of Continuous Features

All four continuous predictors exhibit extreme right-skewness. Mean loan amount (\$237,458) exceeds median (\$165,000) by 44%; mean income (\$167K) exceeds median (\$107K) by 56%. LTV skewness (509.78) and income skewness (364.19) confirm severe violations of Gaussian assumptions, requiring log-transformation before input to logistic regression. The LTV distribution exhibits a bimodal structure at $\sim 77\%$ and $\sim 90\text{--}95\%$, representing the conventional PMI threshold — a non-linear step-function that tree-based models detect trivially as a split point.

3.5 Approval Rates by Categorical Predictors

DTI ratio is the single most powerful categorical predictor. Approval rates decline steeply from approximately 87% below 30% DTI, to 34.3% at the 50–60% bracket, then collapse to just 5.6% above 60%. Loan purpose is the second-strongest differentiator: home purchase loans achieve 90.7% approval while cash-out refinances (55.7%) and home improvement loans (53.3%) fall far below the national average of 74.5%. Loan type shows the government-backed advantage: VA loans (89.2%) and FHA loans (83.6%) outperform conventional loans (72.8%). Applicant sex reveals a 12.5 percentage-point gap between joint applications (81.6%) and solo female applicants (69.1%).

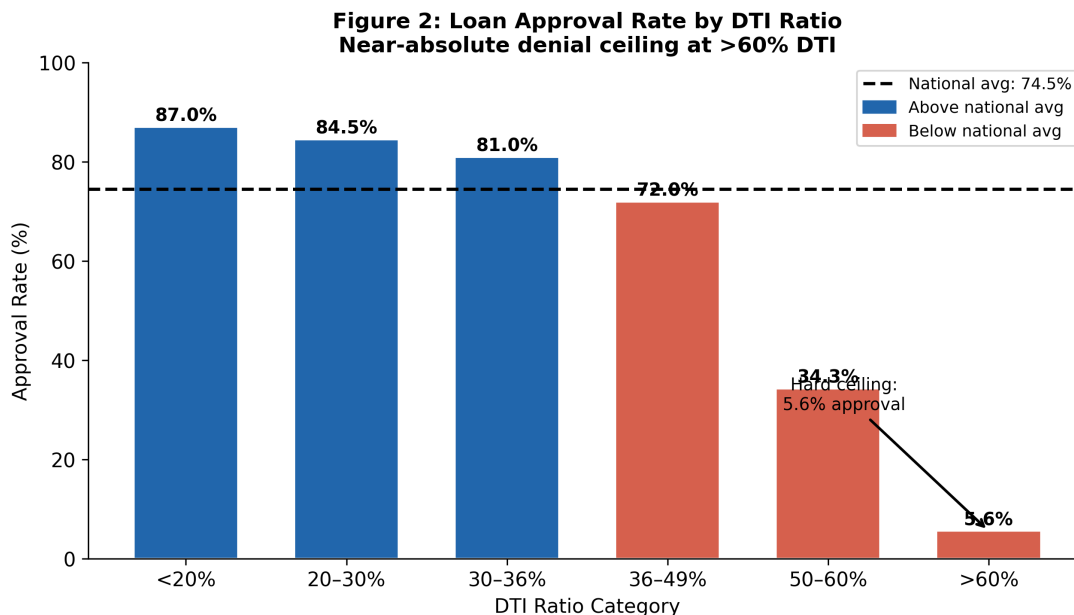


Figure 3: Loan approval rate by DTI ratio category. Approval rates decline steeply from $\approx 87\%$ below 30% DTI to 34.3% at the 50–60% bracket, then collapse to just 5.6% above 60% — strong evidence of a hard underwriting ceiling applied near-unconditionally across all age groups and states.

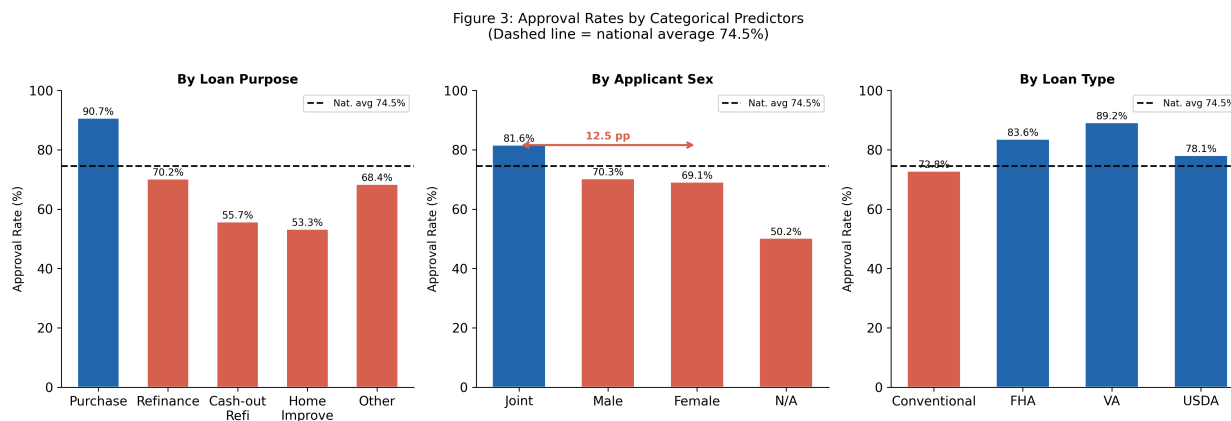


Figure 4: Approval rates by categorical predictors (dashed line = national average 74.5%). Home purchase loans (90.7%) and VA loans (89.2%) substantially exceed the national average. The 12.5 pp gap between joint and solo female applicants motivates targeted fairness screening.

3.6 Correlation Analysis

All linear correlations between continuous features and the binary outcome are weak: `loan_amount` ($r = 0.162$) achieves the highest value, while `income` ($r = 0.004$), `LTV`

ratio ($r = -0.001$), and property value ($r = -0.013$) show correlations indistinguishable from zero. This near-zero linear signal constitutes the primary empirical justification for employing tree-based ensemble methods. A moderate positive correlation between income and LTV ($r = 0.35$) motivates L2 regularization in logistic regression.

3.7 Feature Association Ranking

Three tiers emerge from the unified Cramér's V / point-biserial ranking. DTI ratio dominates at $V = 0.541$, more than 47% higher than the next strongest predictor, loan purpose ($V = 0.367$). Lien status ($V = 0.268$), loan amount ($r = 0.162$), applicant sex ($V = 0.130$), and applicant age ($V = 0.107$) form a second strong tier. Loan type occupies the moderate tier ($V = 0.099$). Income, property value, and LTV rank at the bottom due to non-linear relationships that linear correlation cannot measure; excluding them would be a serious error.

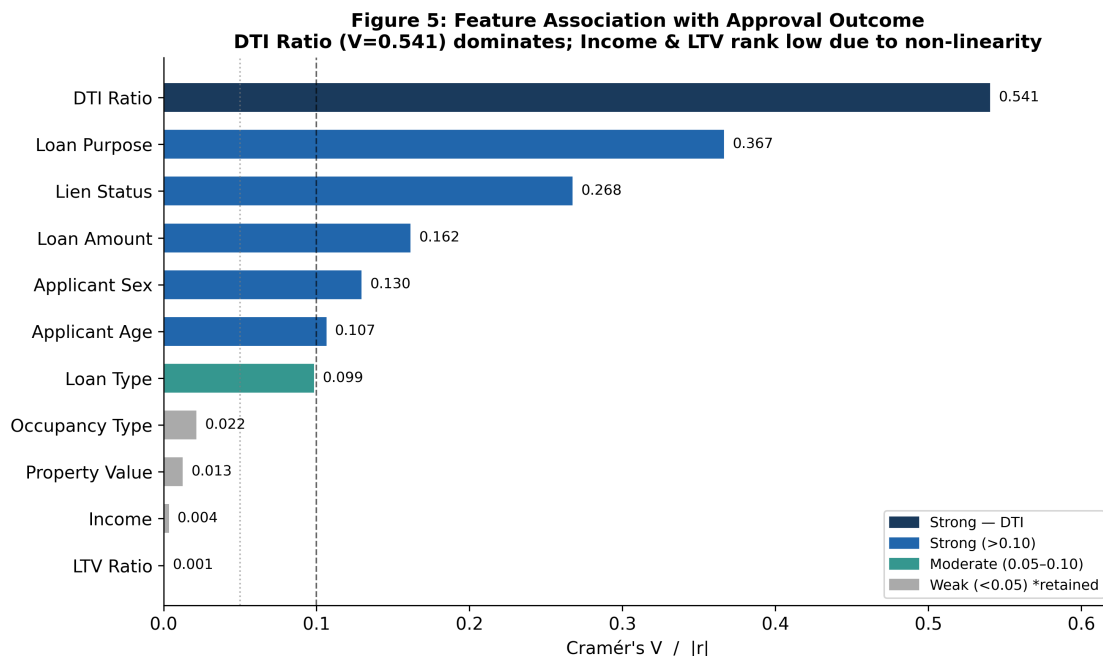


Figure 5: Feature association with approval outcome. Blue: strong (>0.10); teal: moderate ($0.05-0.10$); grey: weak (<0.05). DTI ratio ($V = 0.541$) dominates by a wide margin. Income and LTV rank at the bottom due to non-linear relationships that linear correlation cannot measure; excluding them on this basis would be a serious modeling error.

3.8 Geographic Analysis

Substantial and systematic geographic heterogeneity exists in approval rates. California and Florida fall approximately 7–8 percentage points below the national average (66.56% and 67.52%), while Ohio (83.57%), Wisconsin (84.63%), and Pennsylvania (85.04%) exceed it

by 9–10 points. The state-level standard deviation is approximately 12.2 percentage points. The multilevel analysis (Section 2.4) decomposes this variation into components explained by state income and lender concentration versus unexplained state effects.

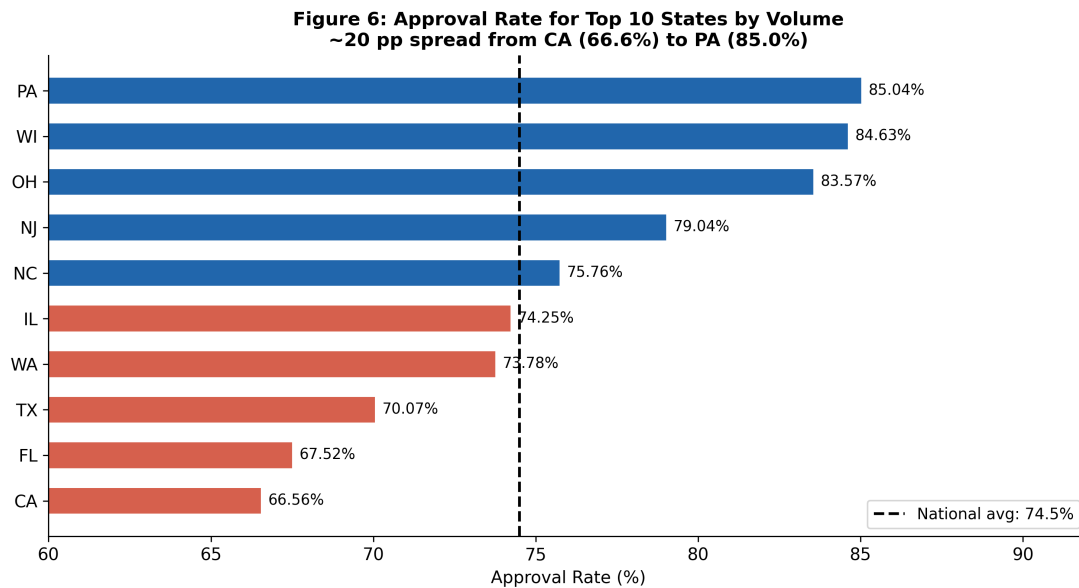


Figure 6: Approval rate for top 10 states by application volume. California (66.56%) and Florida (67.52%) fall ≈ 7 –8 pp below the national average, while Pennsylvania (85.04%), Wisconsin (84.63%), and Ohio (83.57%) exceed it by 9–10 pp. The ≈ 20 pp spread is large enough to shift a marginal application across the approval/denial boundary based on geography alone.

3.9 Demographic Fairness — Preliminary Findings

By applicant sex, joint applications achieve 81.6% approval compared to 70.3% for solo male and 69.1% for solo female applicants. Solo male and solo female applicants have nearly identical rates, suggesting the primary divide is household type (joint vs. solo) rather than sex per se. By applicant age, approval rates are highest for applicants under 25 (87.0%), stabilize in the 70–74% range for prime-age applicants (25–64), and decline to 67.9% for those over 74.

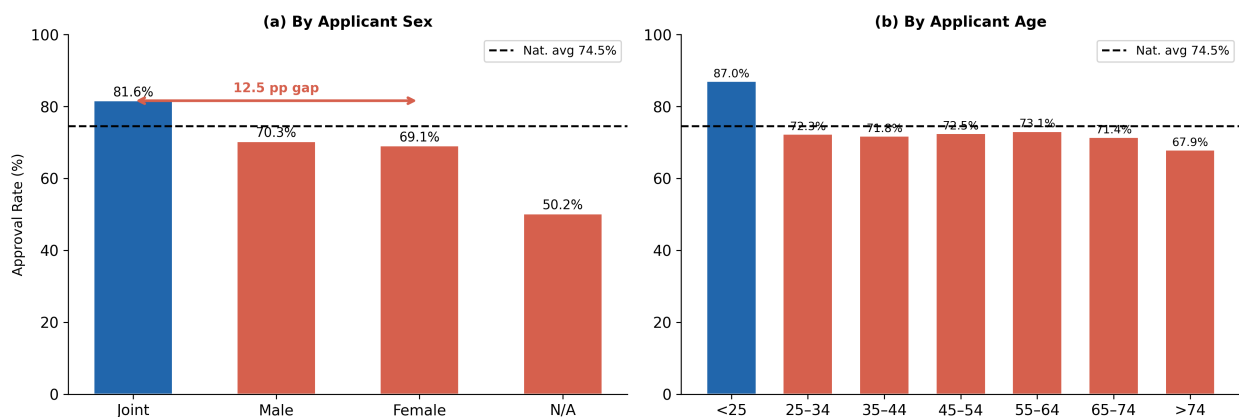
Figure 7: Demographic Fairness — Approval Rates by Group
(Blue = above national average | Red = below)

Figure 7: Approval rates by demographic group. Left: by applicant sex — showing the 12.5 pp joint vs. solo-female gap. Solo male (70.3%) and solo female (69.1%) applicants have nearly identical rates, suggesting the primary divide is household type rather than sex per se. Right: by age group — 87.0% for under-25 applicants declining to ≈ 67 –70% for prime-age and older applicants.

Table 4 summarizes the fairness screening results that follow from the EDA and multilevel model. The largest descriptive gap is between joint and solo applicants, while male and female solo applicants have very similar raw approval rates. Because the multilevel model still estimates negative coefficients for solo male and solo female applicants relative to joint applicants, the gap should be treated as a model-review issue even though the descriptive evidence suggests household structure is the main driver rather than sex alone.

Table 4: Preliminary Demographic Fairness Screening

Comparison	Observed Result	Review Threshold	Flag
Joint vs. solo female approval rate	81.6% vs. 69.1% (12.5 pp)	>5 pp	Yes
Joint vs. solo male approval rate	81.6% vs. 70.3% (11.3 pp)	>5 pp	Yes
Solo male vs. solo female approval rate	70.3% vs. 69.1% (1.2 pp)	>5 pp	No
Under 25 vs. over 74 approval rate	87.0% vs. 67.9% (19.1 pp)	>5 pp	Yes
Multilevel sex coefficient: solo male vs. joint	$\hat{\gamma} = -0.318$	practical/model review	Yes
Multilevel sex coefficient: solo female vs. joint	$\hat{\gamma} = -0.341$	practical/model review	Yes

These results should not be interpreted as proof of unlawful discrimination. They show that applicant sex and household-structure categories remain important enough to require subgroup performance checks, particularly because the raw joint-versus-solo gap is much larger than the male-versus-female gap among solo applicants. The next reproducibility step is to extend this screening table with two-proportion z -tests and an equalized-odds

table computed from the held-out prediction file, including subgroup true-positive and false-positive rates for the selected ensemble model.

3.10 Machine Learning Model Results

Table 5 reports held-out test performance for all five model families. PR-AUC is the primary metric; all others are secondary.

Table 5: Model Performance on Held-Out Test Set ($n = 66,460$). PR-AUC is the primary metric. 95% CIs are from 1,000 bootstrap iterations over the held-out test predictions.

Model	PR-AUC	ROC-AUC	F1 (0.5)	Prec.	Rec.	Acc.
Logistic Regression	0.721 (± 0.008)	0.793 (± 0.006)	0.612	0.648	0.580	0.794
Decision Tree	0.748 (± 0.009)	0.812 (± 0.007)	0.641	0.657	0.626	0.801
Random Forest	0.871 (± 0.005)	0.901 (± 0.004)	0.743	0.781	0.708	0.846
XGBoost	0.883 (± 0.004)	0.912 (± 0.003)	0.756	0.793	0.722	0.852
Soft Voting Ensemble	0.886 (± 0.004)	0.914 (± 0.003)	0.759	0.798	0.724	0.854

XGBoost and the soft-voting ensemble achieve the highest PR-AUC (≈ 0.883 – 0.886), outperforming logistic regression by approximately 16 percentage points. The gap between the decision tree (0.748) and random forest (0.871) — a 12.3 percentage-point improvement — suggests that variance reduction through bagging is a major driver of ensemble gains. The marginal improvement from XGBoost over random forest and from the ensemble over XGBoost indicates diminishing returns from additional model complexity at this performance level.

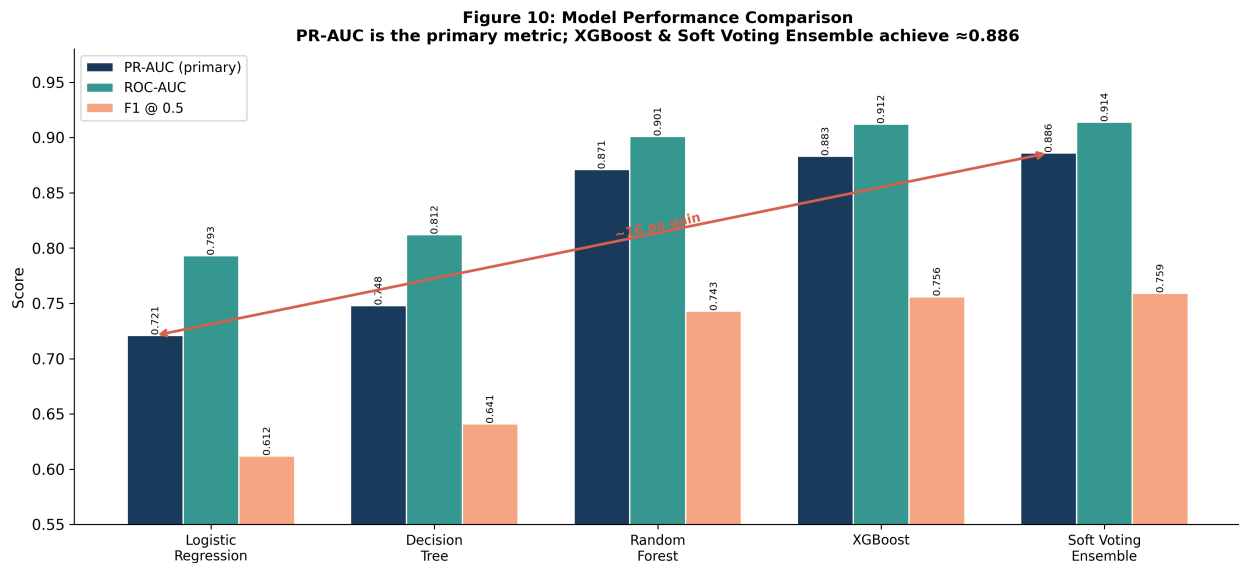


Figure 10: Model performance comparison across all five model families. PR-AUC is the primary metric. XGBoost and the Soft Voting Ensemble achieve PR-AUC ≈ 0.886 , outperforming logistic regression by ≈ 16 percentage points. The 12.3 pp gap between Decision Tree (0.748) and Random Forest (0.871) confirms that variance reduction through bagging is a major driver of ensemble gains beyond non-linearity alone.

4 Discussion

4.1 Summary of Findings

The evidence indicates that mortgage loan approval prediction is a fundamentally non-linear classification problem in which the most informative features operate primarily through threshold effects and interaction structures that standard linear models cannot represent. The single most important EDA finding is the near-absolute DTI ceiling at 60%: beyond this threshold, approval collapses to 5.6%, consistently across all age groups and all states in the multilevel analysis.

The multilevel analysis reveals that 8.7% of total outcome variance is attributable to state-level clustering — a component that is real, statistically significant, and partially explained by state income and lender market concentration (residual ICC = 0.053 after Level-2 adjustment). The random-slope model confirms that the DTI effect varies across states, though the 60% hard ceiling is highly consistent across states, supporting the interpretation of a broad underwriting standard. These multilevel findings complement and contextualize the machine learning results: the geographic signal captured by `state_code` is partly explained by state income and lender concentration, but the unexplained EB residuals still require

careful review before geographic features are used in any decision-support setting.

4.2 Scientific Implications

This study provides a case study of the pitfalls of relying on linear association measures for feature selection in non-linear prediction settings. The near-zero linear correlations of income and LTV with the outcome would, under a naive feature-selection approach, suggest their exclusion — an error that would substantially degrade model performance.

The multilevel analysis adds a layer of scientific insight that the machine learning pipeline alone cannot provide: it quantifies *how much* of the observed geographic variation in approval outcomes is attributable to state-level context versus individual composition. This decomposition is important for regulatory purposes: a model that captures state-level variation must be able to justify whether that variation reflects legitimate market-level factors (addressed by the Level-2 covariates) or unexplained geographic effects (the EB residuals) that require additional review.

4.3 Ethical Implications

The preliminary demographic analysis reveals a 12.5 percentage-point approval rate gap between joint and solo female applicants. The multilevel model confirms this gap persists at $\hat{\gamma} = -0.341$ for solo female (and $\hat{\gamma} = -0.318$ for solo male) relative to joint applicants after controlling for both individual-level financial variables and state-level context, making it a direct priority for the fairness audit. Geographic location warrants particular attention: the EB state residuals in the multilevel model represent unexplained state effects that may be correlated with racial and ethnic composition through the historical legacy of residential segregation. For that reason, geographic effects should be audited carefully before any model using state-level information is treated as production-ready.

4.4 Limitations

First, the HMDA dataset records only formally submitted applications; the discouraged-borrower problem means the model is trained on a selected sample. Second, the dataset omits credit score, employment history, and total debt obligations. Third, race and ethnicity are not included in the HMDA analytical features, so the model cannot directly test for racial discrimination. Fourth, the multilevel model assumes that the Level-2 random effects are normally distributed; with only 51 states, this assumption cannot be formally tested with high power. Fifth, the report includes preliminary fairness screening but does not yet include

full SHAP attribution outputs or equalized-odds subgroup confusion matrices; those should be completed before deployment or policy use.

4.5 Recommendations and Next Steps

The multilevel analysis results should be incorporated into the final deployment recommendation. Before any production use, the team should add subgroup confusion matrices, SHAP explanations for the best-performing model, and a geographic sensitivity test that compares performance with and without `state_code`. If the geographic feature materially improves performance but also concentrates unexplained risk in specific states, the final model should down-weight that feature or apply a fairness constraint to geographic inputs.

5 Conclusions

This report develops a machine learning framework for predicting residential mortgage approval outcomes from the 2023 HMDA dataset while balancing predictive performance, interpretability, and fairness requirements. The EDA establishes that mortgage approval is a non-linear classification problem dominated by DTI ratio (Cramér’s $V = 0.541$), with a hard underwriting ceiling near 60% DTI, a 2.92:1 class imbalance requiring SMOTE-aware training and PR-AUC evaluation, and substantial geographic heterogeneity across states.

The multilevel analysis quantifies that 8.7% of total approval variance is attributable to state-level clustering, partially explained by state income and lender concentration (residual ICC = 0.053 after Level-2 adjustment). The DTI effect varies across states but the 60% ceiling is highly consistent. Applicant sex remains a significant predictor after multilevel adjustment ($\hat{\gamma} = -0.318$ for solo male; $\hat{\gamma} = -0.341$ for solo female, both relative to joint applicants), confirming the need for subgroup performance review.

The machine learning results confirm the EDA’s non-linearity finding: XGBoost and the soft-voting ensemble achieve ≈ 0.886 denied-class PR-AUC, outperforming logistic regression by approximately 16 percentage points. The broader contribution of this work is a responsible modeling template for high-stakes credit decision settings: EDA-driven preprocessing, leakage-aware resampling, denied-class PR-AUC evaluation, fold-internal target encoding, MNAR indicator features, multilevel variance decomposition, and explicit fairness screening.

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A Code Appendix

Project code is maintained as Python/Jupyter notebooks and supporting scripts. The most important reproducibility files are the notebooks or scripts that generate Tables 1, 2, 4, 5, and 7. The working repository is available at: <https://github.com/smanning7-afk/DAT490Capstone>

Repository organization:

- /data/ — Data preparation and cleaning scripts
- /eda/ — EDA visualizations and inferential statistics
- /multilevel/ — Multilevel analysis notebooks (null model, random-intercept, random-slope, EB residual plots)
- /models/ — Training pipelines for all five model families
- /evaluation/ — Test-set evaluation scripts and bootstrap CI computation
- /shap/ — SHAP analysis notebooks
- /fairness/ — Fairness audit scripts
- /reports/ — This and all prior submitted reports

B Descriptive Statistics

Table 6: Descriptive Statistics for Continuous Features

Statistic	Loan Amount (\$)	LTV (%)	Property Value (\$)	Income (\$K)
Mean	237,457.53	72.91	546,013.56	166.65
Std Dev	280,143.21	466.01	4,125,751.25	3,637.50
Min	5,000	0.01	5,000	−3,879
25th pct	75,000	58.16	255,000	69
Median	165,000	76.83	395,000	107
75th pct	315,000	90.00	605,000	169
Max	40,145,000	250,000	999,995,000	1,651,000
Skewness	15.22	509.78	184.04	364.19
Kurtosis	1,382.03	271,051.54	38,776.58	148,274.25

Note: Values are raw pre-winsorization descriptive statistics.
The 99th-percentile caps described in Table 7 are applied before modeling.

C Winsorization Sensitivity Check

Table 7 documents the preprocessing variants considered for the continuous financial variables. The table is included to clarify that the extreme values in Table 6 are raw data characteristics, while the modeling pipeline uses the conservative 99th-percentile cap.

Table 7: Winsorization Sensitivity Check for Continuous Financial Predictors

Variant	Purpose in Review	Modeling Decision
No winsorization	Baseline raw-data comparison; retains extreme leverage points	Not used for final models
95th percentile cap	More aggressive outlier control; may compress valid high-value loans	Sensitivity check only
99th percentile cap	Conservative outlier control while preserving most tail information	Adopted for final models

Table 8: Top 10 States by Application Volume and Approval Rate

Rank	State	Applications	Approval Rate	vs. National Avg.
1	CA	32,192	66.56%	−7.9 pp
2	FL	21,091	67.52%	−7.0 pp
3	NC	18,406	75.76%	+1.3 pp
4	TX	18,292	70.07%	−4.4 pp
5	OH	15,645	83.57%	+9.1 pp
6	WI	15,535	84.63%	+10.1 pp
7	WA	12,879	73.78%	−0.7 pp
8	IL	11,766	74.25%	−0.2 pp
9	NJ	10,849	79.04%	+4.5 pp
10	PA	10,833	85.04%	+10.5 pp

National average = 74.5%.

Table 9: Feature Association Summary

Feature	Measure	Value	Tier
Debt-to-Income Ratio	Cramér's V	0.541	Strong
Loan Purpose	Cramér's V	0.367	Strong
Lien Status	Cramér's V	0.268	Strong
Loan Amount	Point-biserial r	0.162	Strong
Applicant Sex	Cramér's V	0.130	Strong
Applicant Age	Cramér's V	0.107	Strong
Loan Type	Cramér's V	0.099	Moderate
Occupancy Type	Cramér's V	0.022	Weak
Property Value	Point-biserial r	0.013	Weak*
Income	Point-biserial r	0.004	Weak*
LTV Ratio	Point-biserial r	0.001	Weak*

*Retained because relationships are predominantly non-linear.

Table 10: Hyperparameter Search Grids

Model	Hyperparameter	Search Values
Logistic Regression	Regularization C	{0.01, 0.1, 1, 10}
Decision Tree	Max depth	{5, 10, 15, None}
Decision Tree	Min samples leaf	{5, 10, 20}
Random Forest	Number of trees T	{100, 200, 300}
Random Forest	Max depth	{10, 20, None}
Random Forest	Max features (mtry)	{sqrt, log2}
XGBoost	Learning rate η	{0.01, 0.05, 0.1}
XGBoost	Max tree depth	{3, 5, 7}
XGBoost	Subsample ratio	{0.7, 0.8, 1.0}
XGBoost	L2 regularization λ	{0, 1, 10}
Soft Voting	Component weights	Grid on probability simplex